Definition 3.1

The radian measure of an angle is the ratio of the arc length it subtends, to the radius of the circle in which it is the central angle.

Consider a circle of radius r. Let s be the arc length subtending an angle θ at the centre.

Then, $\theta = \dfrac{\text{arc length}}{\text{radius}} = \dfrac{s}{r}$ radians. Hence, $s = r\theta$.

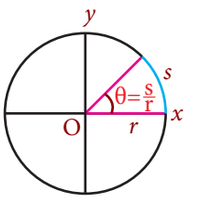


Figure 3.5

(i) All circles are similar. Thus, for a given central angle in any circle, the ratio of the intercepted arc length to the radius is always constant.

(ii) When $s = r$, we have an angle of 1 radian. Thus, one radian is the angle made at the centre of a circle by an arc with length equal to the radius of the circle.

(iii) Since the lengths s and r have same unit, θ is unitless and thus, we do not use any notation to denote radians.

(iv) $\theta = 1$ radian measure, if $s = r$
$\theta = 2$ radian measure, if $s = 2r$
Thus, in general $\theta = k$ radian measure, if $s = kr$.
Hence, radian measure of an angle tells us how many radius lengths, we need to sweep out along the circle to subtend the angle θ.

(v) Radian angle measurement can be related to the edge of the unit circle. In radian system, we measure an angle by measuring the distance travelled along the edge of the unit circle to where the terminal side of the angle intersect the unit circle .

3.3.1 Relationship between Degree and Radian Measures

We have degree and radian units to measure angles. One measuring unit is better than another if it can be defined in a simpler and more intuitive way. For example, in measuring temperature, Celsius unit is better than Fahrenheit as Celsius was defined using $0°$ and $100°$ for freezing and boiling points of water. Radian measure is better for conversion and calculations. Radian measure is more convenient for analysis whereas degree measure of an angle is more convenient to communicate the concept between people. Greek Mathematicians observed the relation of π which arises from circumference of a circle and thus, π plays a crucial role in radian measure.

In unit circle, a full rotation corresponds to $360°$ whereas, a full rotation is related to 2π radians, the circumference of the unit circle. Thus, we have the following relations:

$$2\pi \text{ radians } = 360°, \text{ which reduces to } \pi \text{ radians } = 180°.$$

$$\text{Thus, } 1 \text{ radian } = \left(\frac{180}{\pi}\right)^\circ \quad \text{or} \quad 1° = \frac{\pi}{180} \text{ radians.}$$

$$x \text{ radian } = \left(\frac{180x}{\pi}\right)^\circ \quad \text{or} \quad x° = \frac{\pi x}{180} \text{ radians.}$$

Observe that the scale used in radians is much smaller than the scale in degrees. The smaller scale makes the graphs of trigonometric functions more visible and usable. The above relation gives a way to convert radians into degrees or degrees into radians.

(i) The ratio of the circumference of any circle to its diameter is always a constant. This constant is denoted by the irrational number π.

(ii) Mark a point P on a unit circle and put the unit circle on the number line so that P touches the number 0. Allow the circle to roll along the number line. The point P will